In [34]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [142...]:
```
df
```

Out[142...]

	Jersey No	Player	Matches	Inns	Not Out	Runs	Highest Score	Avg	Balls faced	Strike rate	50s
0	1	KL Rahul	14	14	2	670	132*	55.83	518	129.34	1
1	2	Shikhar Dhawan	17	17	3	618	106*	44.14	427	144.73	2
2	3	David Warner	16	16	2	548	85*	39.14	407	134.64	0
3	4	Shreyas Iyer	17	17	2	519	88*	34.60	421	123.27	0
4	5	Ishan Kishan	14	13	4	516	99	57.33	354	145.76	0
...	...	...	...	...	...	...	...	...	...	...	...
128	129	Khaleel Ahmed	7	1	0	0	0*	0.00	2	0.00	0
129	130	Arshdeep Singh	8	1	0	0	0*	0.00	3	0.00	0
130	131	Daniel Sams	3	1	0	0	0*	0.00	2	0.00	0
131	132	Shreevats Goswami	2	2	0	0	0*	0.00	4	0.00	0
132	133	Trent Boult	15	1	0	0	0*	0.00	1	0.00	0

133 rows × 18 columns

Load the dataset

In [3]:
```
file_path = "C:/Users/praga/Downloads/IPL Data Set.xlsm"
xls = pd.ExcelFile(file_path)
df = pd.read_excel(xls, sheet_name="IPL_Dataset")
```

Clean column names

In [144...]:
```
# Data cleaning with the columns names and mapping values
df.columns = df.columns.str.strip()
df.rename(columns={df.columns[10]: "100s", df.columns[11]: "50s"}, inplace=True)
print(df.columns)
```

```
Index(['Jersey No', 'Player', 'Matches', 'Inns', 'Not Out', 'Runs',
       'Highest Score', 'Avg', 'Balls faced', 'Strike rate', '100s', '50s',
       '6s', 'First_Name', 'Last_Name', 'Cleaned_Highest_score',
       'Cleaned_Highest_Score', 'Boundary_Percentage'],
      dtype='object')
```

Q1. What is the maximum number of matches played by an individual player in a season? Print the player name along with the number of matched played.

In [8]:
```
max_matches = df["Matches"].max()
player_max_matches = df[df["Matches"] == max_matches][["Player", "Matches"]]
print("Q1 Answer:")
print(player_max_matches)
```

```
Q1 Answer:
            Player  Matches
1   Shikhar Dhawan       17
3     Shreyas Iyer       17
19  Marcus Stoinis       17
67   Kagiso Rabada       17
```

In [125…
```
print(df.columns)
```

```
Index(['Jersey No', 'Player', 'Matches', 'Inns', 'Not Out', 'Runs',
       'Highest Score', 'Avg', 'Balls faced', 'Strike rate', '50s', '4s', '6s',
       'First_Name', 'Last_Name', 'Cleaned_Highest_score',
       'Cleaned_Highest_Score', 'Boundary_Percentage'],
      dtype='object')
```

Q2. Top 2 players with maximum Average who have scored atleast 2 half centuries ?

In [148…
```
qualified_players = df[df["50s"] >= 2]
top_avg_players = qualified_players.nlargest(2, "Avg")[["Player", "Avg"]]
print("Q2 Answer:")
print(top_avg_players)
```

```
Q2 Answer:
          Player    Avg
57  Deepak Hooda  101.0
60    Tom Curran   83.0
```

Q.3 Create 2 new columns based on Player name. First column will have first name and second column will have last name. Eg: for the player Shikhar Dhawan, Shikhar will be the first name and Dhawan will be the last name

In [69]:
```
# Ensure "Player" column is a string
df["Player"] = df["Player"].astype(str)

# Split player names into first and last names safely
df[["First_Name", "Last_Name"]] = df["Player"].str.split(n=1, expand=True)

# Check for missing values
print(df[["Player", "First_Name", "Last_Name"]].head())
```

```
           Player First_Name Last_Name
0       KL Rahul         KL     Rahul
1  Shikhar Dhawan    Shikhar    Dhawan
2    David Warner      David    Warner
3    Shreyas Iyer    Shreyas      Iyer
4    Ishan Kishan      Ishan    Kishan
```

Q4. Create a new column (Cleaned_Highest_score) based on Highest score variable. Remove the Asterik(*) mark and convert the data type into INT.

In [67]:
```
# Ensure "Highest Score" is a string and remove "*" properly
df["Cleaned_Highest_Score"] = df["Highest Score"].astype(str).str.replace(r"\*",

# Convert to numeric (handling NaN values)
df["Cleaned_Highest_Score"] = pd.to_numeric(df["Cleaned_Highest_Score"], errors=
```

In [54]:
```
print(df[["Player", "Cleaned_Highest_score"]].head())
```

```
           Player  Cleaned_Highest_score
0       KL Rahul                    132
1  Shikhar Dhawan                    106
2    David Warner                     85
3    Shreyas Iyer                     88
4    Ishan Kishan                     99
```

Q5. Print the total number of centuries scored in the entire season

In [57]:
```
# Ensure "Runs" column is numeric
df["Runs"] = pd.to_numeric(df["Runs"], errors="coerce").fillna(0).astype(int)

# Count the number of innings where a player scored at least 100 runs
total_centuries = (df["Runs"] >= 100).sum()

print("Total number of centuries scored in the entire season:", total_centuries)
```

```
Total number of centuries scored in the entire season: 58
```

Q6. Print all the player names whose strike rate is less than the average strike rate of all players in entire season. Print the player name, his strike rate and average strike rate.

In [60]:
```
average_strike_rate = df["Strike rate"].mean()
low_sr_players = df[df["Strike rate"] < average_strike_rate][["Player", "Strike
print("Q6 Answer:")
print(low_sr_players)
```

Q6 Answer:

	Player	Strike rate
51	Ajinkya Rahane	105.60
55	Glenn Maxwell	101.88
58	Vijay Shankar	101.04
61	Josh Philippe	101.29
62	Gurkeerat Singh	88.75
65	Kedar Jadhav	93.93
70	Yashasvi Jaiswal	90.90
71	Shreyas Gopal	94.87
77	Murali Vijay	74.41
79	Chris Jordan	93.54
80	Navdeep Saini	100.00
82	Kamlesh Nagarkoti	70.96
84	Harshal Patel	87.50
85	Jimmy Neesham	105.55
86	Tom Banton	90.00
89	Prabhsimran Singh	100.00
92	Kuldeep Yadav	61.90
94	Moeen Ali	75.00
95	Sandeep Sharma	80.00
96	Shardul Thakur	57.14
98	Rinku Singh	100.00
99	Shivam Mavi	71.42
100	Varun Chakaravarthy	66.66
101	Jaydev Unadkat	69.23
102	Ankit Rajpoot	90.00
104	Shahbaz Nadeem	87.50
105	Pravin Dubey	53.84
107	Deepak Chahar	58.33
108	Ravi Bishnoi	58.33
109	Andrew Tye	100.00
111	Kartik Tyagi	66.66
112	Murugan Ashwin	100.00
114	T Natarajan	60.00
115	Prasidh Krishna	50.00
116	Rahul Chahar	50.00
117	Mohammad Shami	66.66
118	Nikhil Naik	33.33
119	Mujeeb Ur Rahman	33.33
120	Dale Steyn	50.00
121	Varun Aaron	10.00
122	Shahbaz Ahmed	100.00
123	Yuzvendra Chahal	33.33
124	Mitchell Marsh	0.00
125	Umesh Yadav	0.00
126	Bhuvneshwar Kumar	0.00
127	Sheldon Cottrell	0.00
128	Khaleel Ahmed	0.00
129	Arshdeep Singh	0.00
130	Daniel Sams	0.00
131	Shreevats Goswami	0.00
132	Trent Boult	0.00

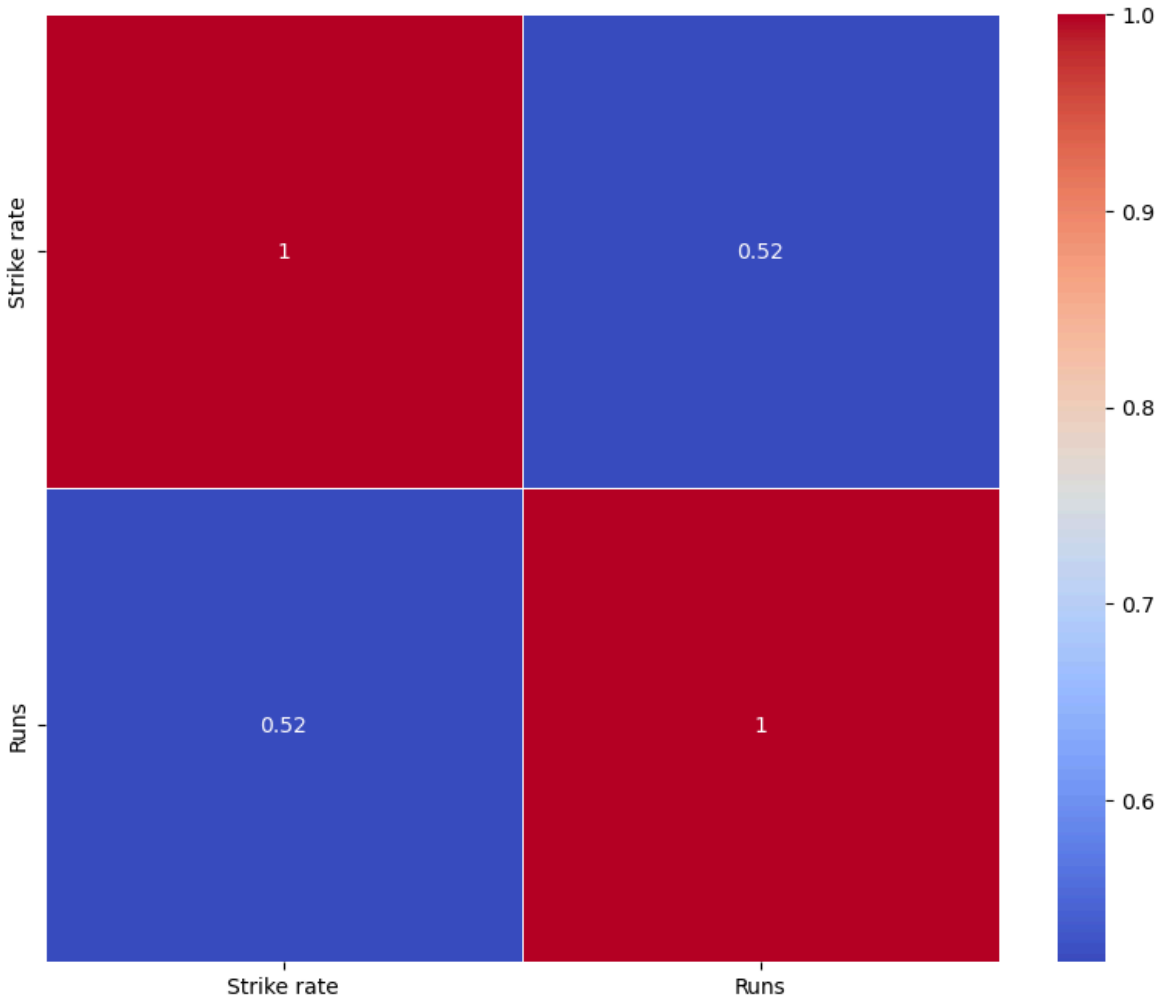
Q7. Please check the correlation between the features and create a heat map.

In [152…

```
correlation_matrix = df[['Strike rate', 'Runs']].corr()
# Create a heatmap
```

```
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', linewidths=0.5)
```

Out[152… <Axes: >



Q8. Check the list of players who has an average greater than 50 as well strike rate above 120. Print player name, average and strike rate.

In [76]:
```
high_perf_players = df[(df["Avg"] > 50) & (df["Strike rate"] > 120)][["Player",
print("Q8 Answer:")
print(high_perf_players)
```

```
Q8 Answer:
             Player     Avg  Strike rate
0          KL Rahul   55.83       129.34
4      Ishan Kishan   57.33       145.76
31   Kieron Pollard   53.60       191.42
36  Wriddhiman Saha   71.33       139.86
37  Ruturaj Gaikwad   51.00       120.71
57     Deepak Hooda  101.00       142.25
60        Tom Curran   83.00       133.87
```

Q9. Please check the list of players who has an average greater than 40 and balls faced above 100. Print player name, average and balls faced.

In [81]:
```
experienced_players = df[(df["Avg"] > 40) & (df["Balls faced"] > 100)][["Player"
print("Q9 Answer:")
print(experienced_players)
```

```
Q9 Answer:
             Player    Avg  Balls faced
0          KL Rahul  55.83          518
1    Shikhar Dhawan  44.14          427
4      Ishan Kishan  57.33          354
8       Virat Kohli  42.36          384
9      ABD Villiers  45.40          286
10    Faf Duplessis  40.81          319
14      Eoin Morgan  41.80          302
24  Kane Williamson  45.28          237
27      Chris Gayle  41.14          210
28       Ben Stokes  40.71          200
31   Kieron Pollard  53.60          140
32    Rahul Tewatia  42.50          183
33  Ravindra Jadeja  46.40          135
36  Wriddhiman Saha  71.33          153
37  Ruturaj Gaikwad  51.00          169
```

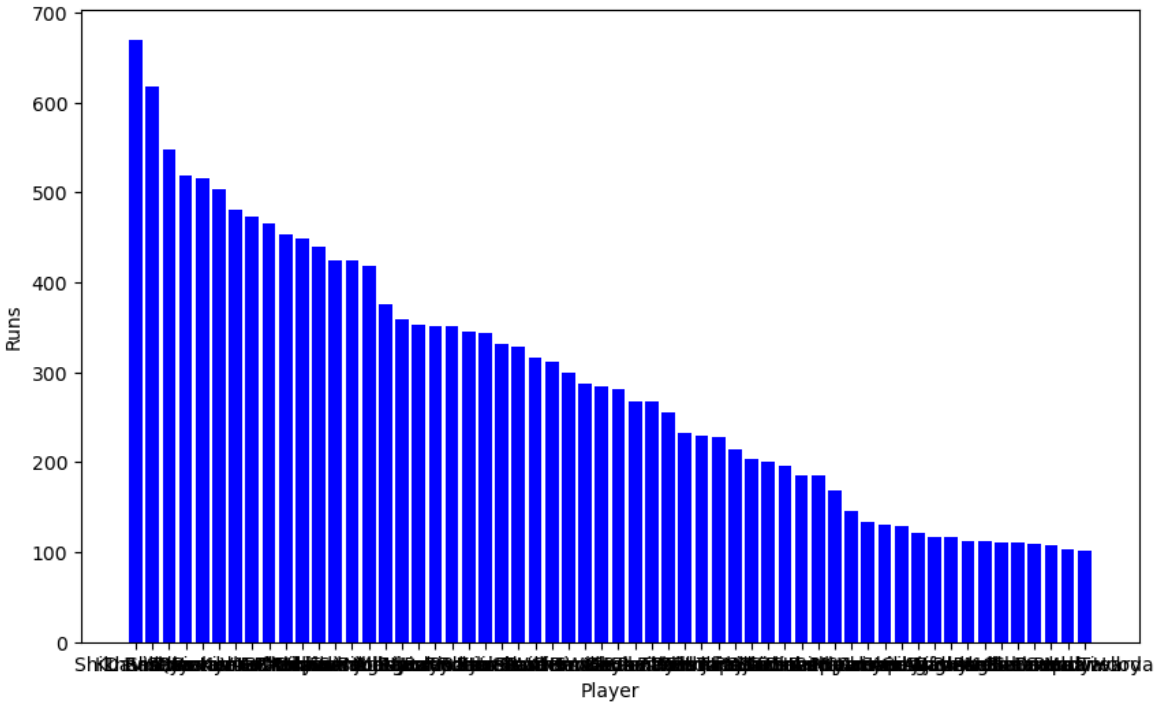
Q10. Players who scored atleast one century in this season. Create visualization.

In [86]:
```
# Ensure "Runs" column is numeric
df["Runs"] = pd.to_numeric(df["Runs"], errors="coerce").fillna(0).astype(int)

# Filter players who have scored at least one century
century_players = df[df["Runs"] >= 100]

# Visualization: Players who scored at least one century
plt.figure(figsize=(10, 6))
plt.bar(century_players["Player"], century_players["Runs"], color="blue")
plt.xlabel("Player")
plt.ylabel("Runs")
```

Out[86]:
```
Text(0, 0.5, 'Runs')
```



Q11. Players who scored atleast 4 half centuries in this season.

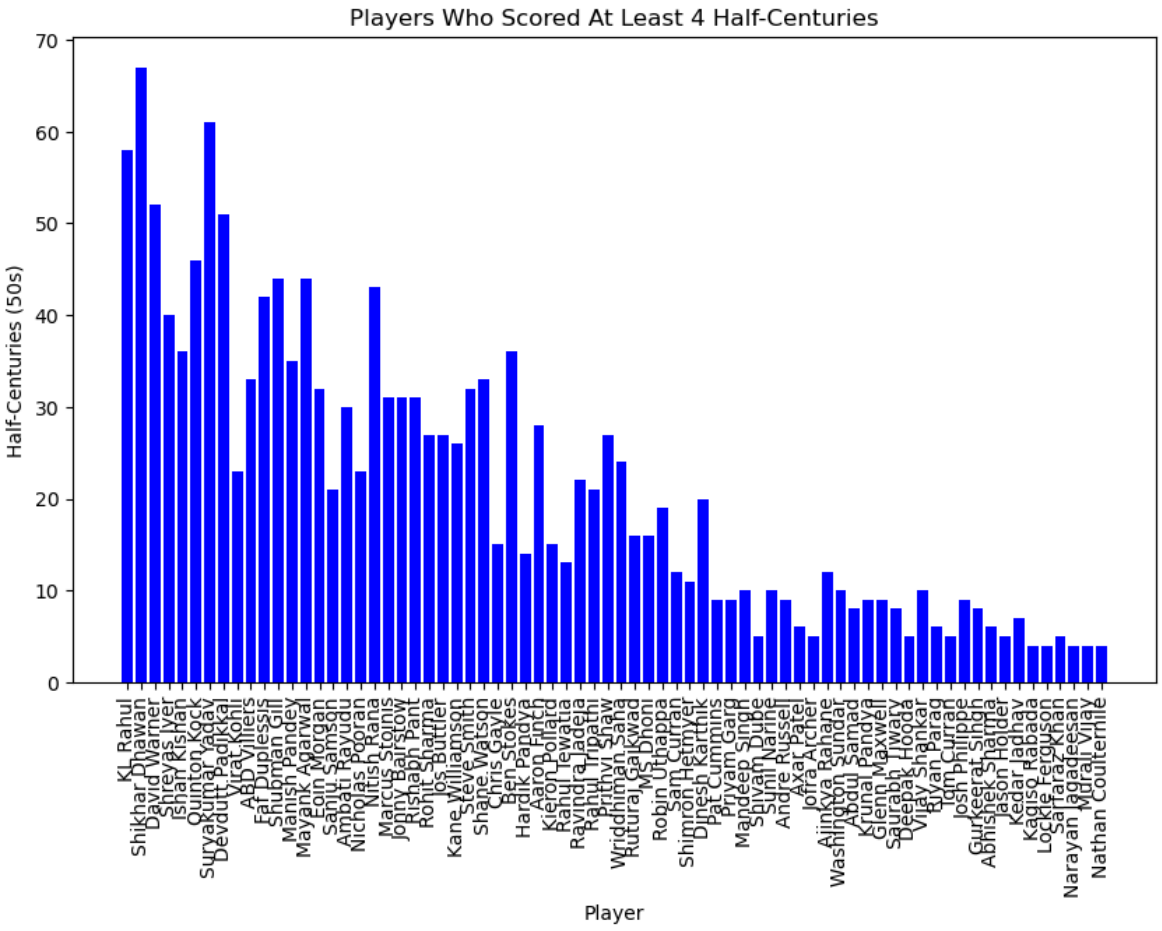
In [91]:
```
print(df.columns[df.columns.duplicated()])
```

```
Index(['50s'], dtype='object')
```

In [93]:
```
df = df.loc[:, ~df.columns.duplicated()]
```

In [162…
```
# Ensure "50s" column is numeric
df["50s"] = pd.to_numeric(df["50s"], errors="coerce").fillna(0).astype(int)

half_century_players = df[df["50s"] >= 4]
plt.figure(figsize=(10, 6))
plt.bar(half_century_players["Player"], half_century_players["50s"], color="blue
plt.xlabel("Player")
plt.ylabel("Half-Centuries (50s)")
plt.title("Players Who Scored At Least 4 Half-Centuries")
plt.xticks(rotation=90)
plt.show()
print("Players who scored at least 4 half-centuries in the season:")
print(half_century_players[["Player", "50s"]])
```



```
Players who scored at least 4 half-centuries in the season:
                Player  50s
0            KL Rahul   58
1       Shikhar Dhawan   67
2         David Warner   52
3         Shreyas Iyer   40
4         Ishan Kishan   36
..                 ...  ...
68     Lockie Ferguson    4
75       Sarfaraz Khan    5
76  Narayan Jagadeesan    4
77        Murali Vijay    4
81  Nathan Coulternile    4

[72 rows x 2 columns]
```

Q12. Check the list of players who hit more than 45 boundaries and more than 10 sixes in this season.

In [100…

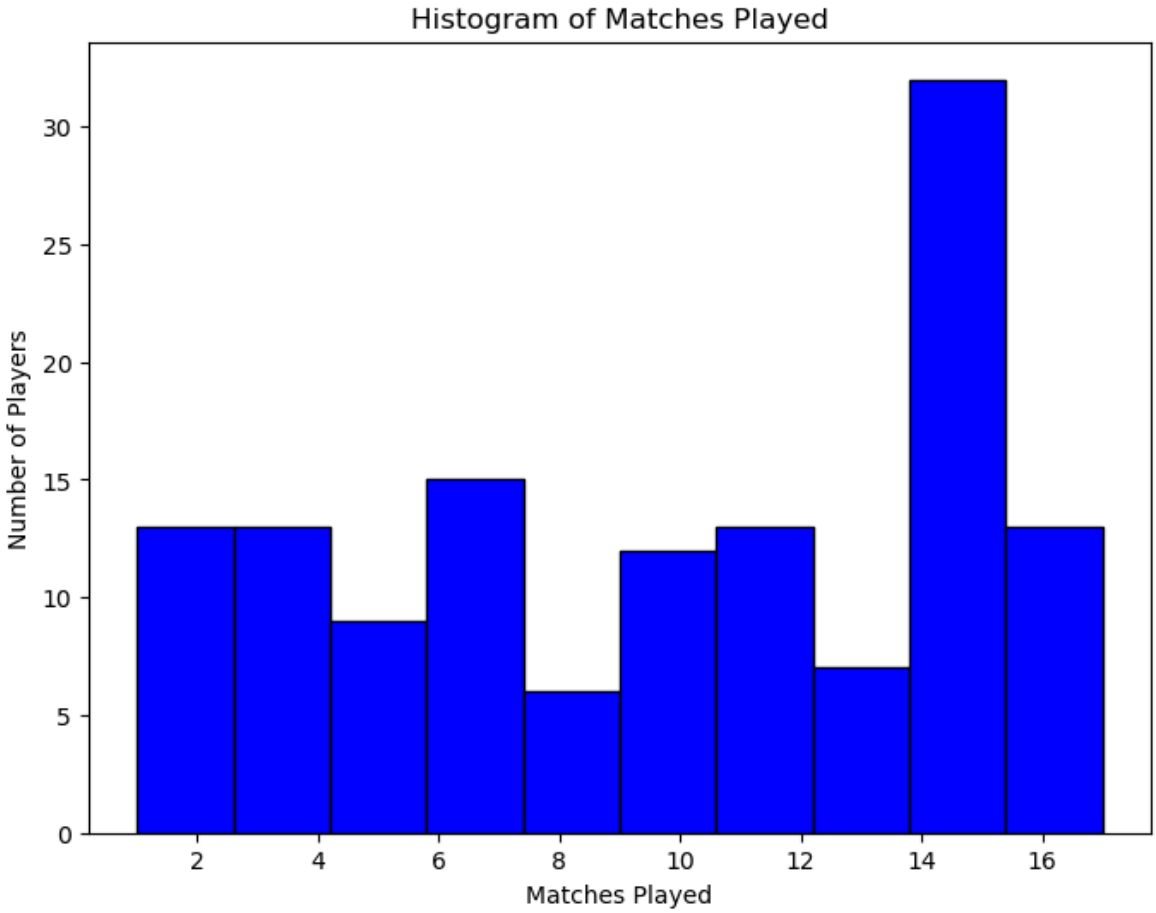
```
boundary_hitters = df[(df["4s"] > 45) & (df["6s"] > 10)][["Player", "4s", "6s"]]
print("Q12 Answer:")
print(boundary_hitters)
```

```
Q12 Answer:
             Player  4s  6s
0          KL Rahul  58  23
1     Shikhar Dhawan  67  12
2       David Warner  52  14
5       Quinton Kock  46  22
6  Suryakumar Yadav  61  11
```

Q13. Plot a histogram of number of matches played in a season by players.

In [103…

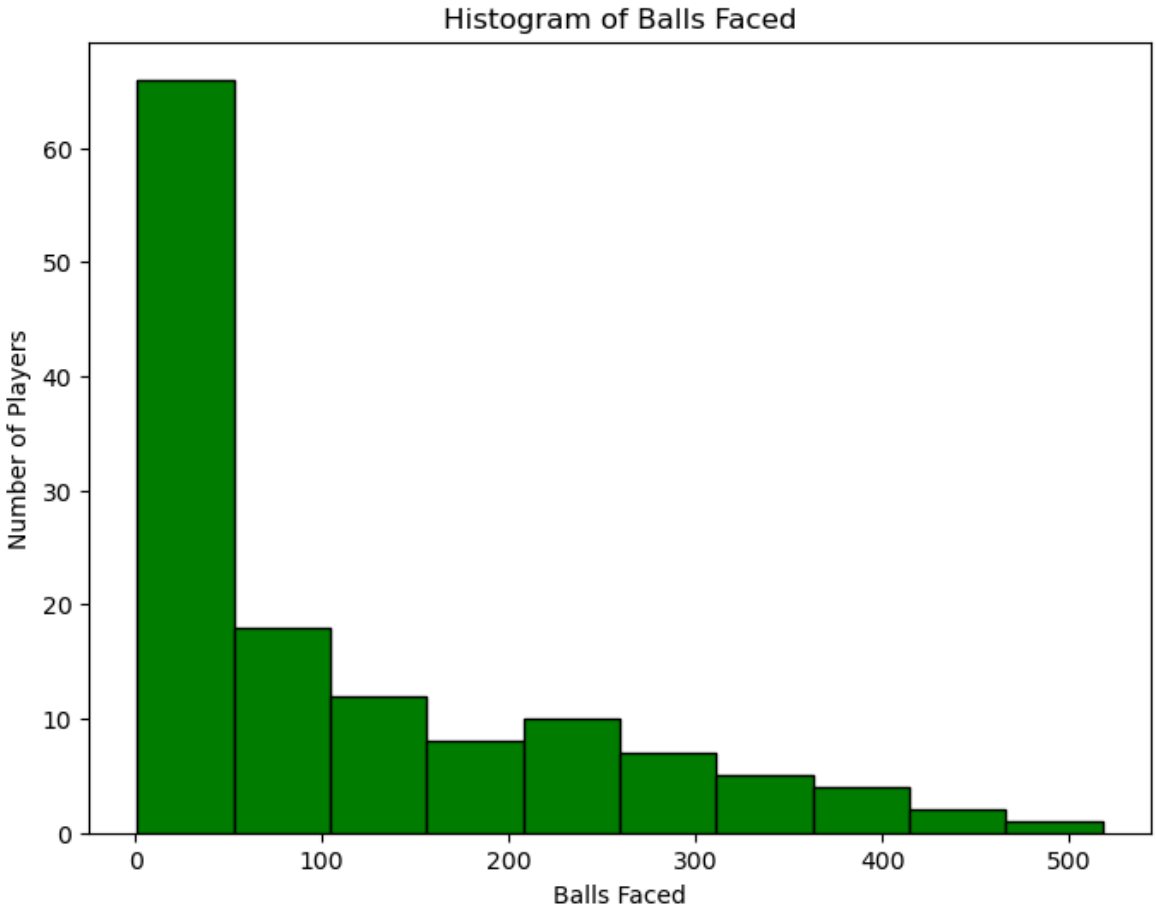
```
plt.figure(figsize=(8,6))
plt.hist(df["Matches"], bins=10, color='blue', edgecolor='black')
plt.xlabel("Matches Played")
plt.ylabel("Number of Players")
plt.title("Histogram of Matches Played")
plt.show()
```



Q14. Plot the histogram of balls faced by players

In [109…

```
plt.figure(figsize=(8,6))
plt.hist(df["Balls faced"], bins=10, color='green', edgecolor='black')
plt.xlabel("Balls Faced")
plt.ylabel("Number of Players")
plt.title("Histogram of Balls Faced")
plt.show()
```



Q15. Top 10 players with most runs in a season.

In [112…
```
top_run_scorers = df.nlargest(10, "Runs")[['Player', 'Runs']]
print("Q15 Answer:")
print(top_run_scorers)
```

```
Q15 Answer:
             Player  Runs
0          KL Rahul   670
1    Shikhar Dhawan   618
2      David Warner   548
3      Shreyas Iyer   519
4      Ishan Kishan   516
5      Quinton Kock   503
6  Suryakumar Yadav   480
7  Devdutt Padikkal   473
8       Virat Kohli   466
9      ABD Villiers   454
```

Q16. Print the players who played the match but didn't get the batting.

In [164…
```
players_no_batting = df[df["Inns"] == 0][["Player", "Matches"]]
print("Q16 Answer:")
print(players_no_batting)
```

```
Q16 Answer:
Empty DataFrame
Columns: [Player, Matches]
Index: []
```

Q17. Create a new column to show the percentage of total runs scored in 4s and 6s. Then print the top 5 players with maximum percentage

In [118…
```
df["Boundary_Percentage"] = ((df["4s"] * 4 + df["6s"] * 6) / df["Runs"]) * 100
top_boundary_players = df.nlargest(5, "Boundary_Percentage")[['Player', 'Boundar
print("Q17 Answer:")
print(top_boundary_players)
```

```
Q17 Answer:
            Player  Boundary_Percentage
109     Andrew Tye           100.000000
48   Andre Russell            76.923077
74    Chris Morris            76.470588
29   Hardik Pandya            73.309609
47    Sunil Narine            72.727273
```

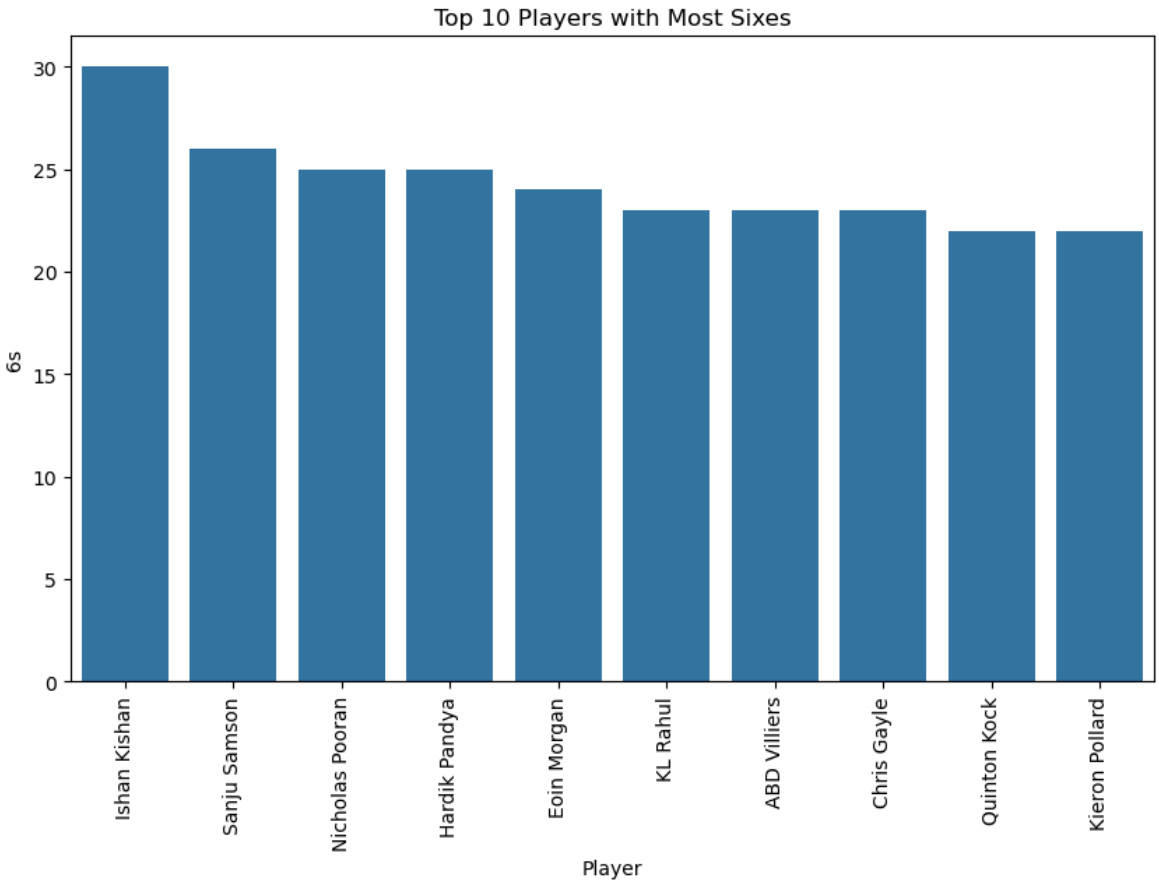
Q18. Print the players with top 5 Not out percentages (Not Out percentage can be calculated as number of Not outs divided by Innings).

In [170…
```
df["NotOut_Percentage"] = (df["Not Out"] / df["Inns"]) * 100
top_not_outs = df.nlargest(5, "NotOut_Percentage")[['Player', 'NotOut_Percentage
print("Q18 Answer:")
print(top_not_outs)
```

```
Q18 Answer:
              Player  NotOut_Percentage
62   Gurkeerat Singh              100.0
68   Lockie Ferguson              100.0
93       Imran Tahir              100.0
97     Mohammad Nabi              100.0
105     Pravin Dubey              100.0
```

Q19. Create visualization of top 10 players with highest number of sixes.

In [173…
```
top_six_hitters = df.nlargest(10, "6s")
plt.figure(figsize=(10,6))
sns.barplot(x='Player', y='6s', data=top_six_hitters)
plt.xticks(rotation=90)
plt.title("Top 10 Players with Most Sixes")
plt.show()
```

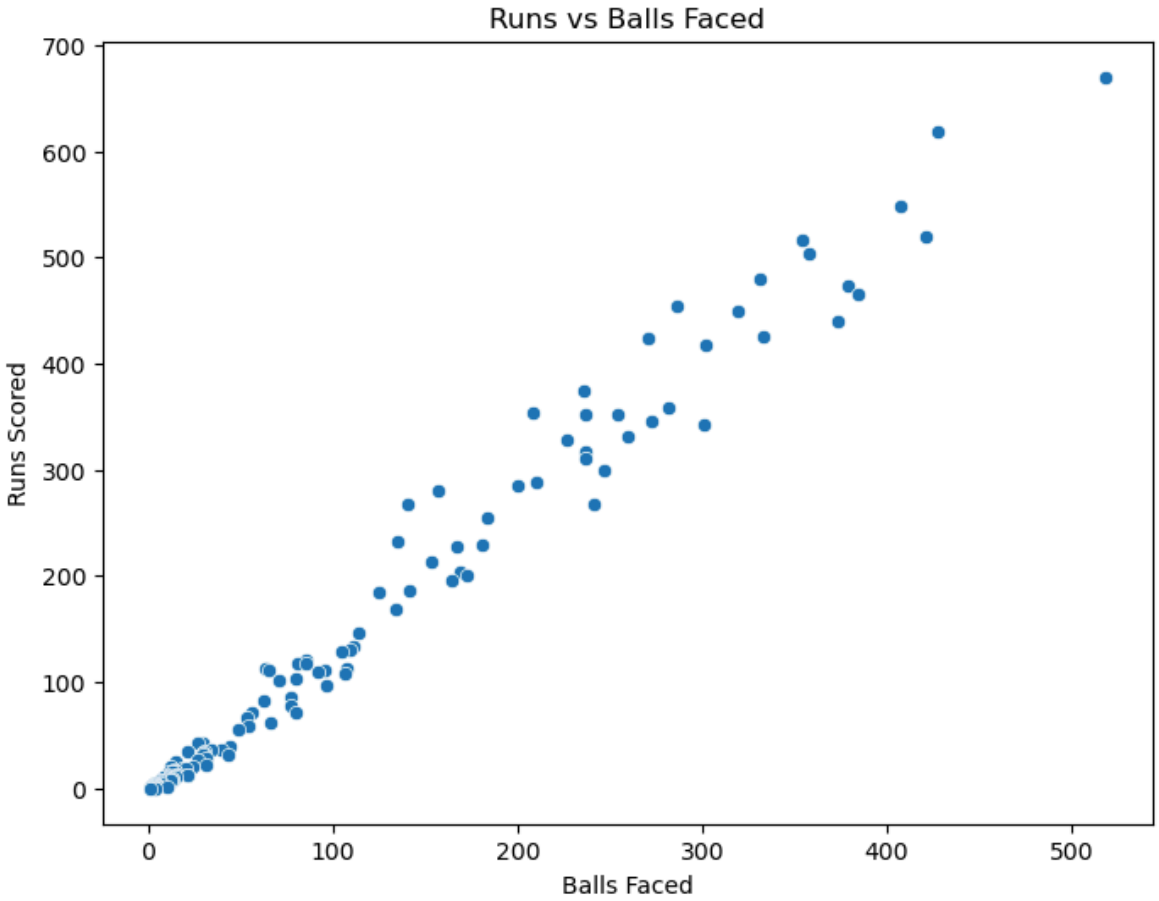


Q20. Scatter plot of runs scored by a player v/s balls faced in a season. Then find the relationship between these 2 variables.

In [178…

```
plt.figure(figsize=(8,6))
sns.scatterplot(x=df['Balls faced'], y=df['Runs'])
plt.xlabel("Balls Faced")
plt.ylabel("Runs Scored")
plt.title("Runs vs Balls Faced")
plt.show()

# Relationship between Runs and Balls Faced
correlation = df[['Balls faced', 'Runs']].corr().iloc[0,1]
print("Q20 Answer: Correlation between Runs and Balls Faced:", correlation)
```




```
Q20 Answer: Correlation between Runs and Balls Faced: 0.9899480233860832
```

In [ ]: